

When Does Search Beat Reasoning? A Topology-Driven Diagnostic of A* versus Chain-of-Thought

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[Author Names]

[Affiliation]

Abstract

Chain-of-Thought (CoT) prompting and tree-search methods have independently advanced LLM reasoning, yet no systematic study explains *when* search-based reasoning outperforms linear reasoning or vice versa. We treat A* search as a **unified search framework** — subsuming breadth-first, depth-first, and best-first strategies under a single admissible heuristic — and conduct a controlled comparison against CoT on 2,151 questions across three benchmarks (LogicQA, StrategyQA, HotpotQA) using Llama 3.1-8B. Aggregate results are equivocal: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). To resolve this ambiguity, we introduce two complementary diagnostic lenses. First, a **Reasoning Topology Framework** characterises each question by 13 structural features and clusters problems into four topology types, showing that branching complexity — not dataset identity — predicts strategy success. Second, **cross-dataset topic modelling** (NMF, 8 categories) reveals that A* dominates in 5 categories covering **81.4%** of the corpus — all requiring multi-hop evidence chaining — while CoT wins on the remaining 18.6% of single-step inference tasks. A topology-aware router that sees *zero model outputs* recovers 26.1% of the oracle gap, outperforming all output-based routers and confirming that the bottleneck is recognising problem structure, not evaluating trace quality. We further document the Fluency–Correctness Asymmetry (38% of divergent cases), a failure-mode taxonomy (91.1% overthinking), and a $3.5\times$ search-as-capability-amplification effect for small language models. Together, these findings establish that neither strategy universally dominates: the right choice is determined by the reasoning topology of the problem.

1 Introduction

The emergence of Chain-of-Thought (CoT) prompting (Wei et al., 2022) demonstrated that large language models reason better when they externalise intermediate steps. Subsequent work extended this insight from linear chains to richer structures: Tree-of-Thoughts (ToT; Yao et al., 2024) explores alternative reasoning paths via breadth-first or depth-first search, Graph-of-Thoughts (GoT; Besta et al., 2024) allows arbitrary reasoning topologies, and MCTS-based approaches (Hao et al., 2023) apply Monte Carlo rollouts to reasoning trees. These advances raise a fundamental question that remains unanswered:

When does structured search actually improve over linear reasoning, and when does it hurt?

Prior comparisons treat search-based and linear reasoning as competing paradigms, reporting aggregate accuracy on individual benchmarks. But aggregates conceal systematic structure. A method that wins by 10 points on commonsense reasoning and loses by 1 point on formal logic looks equivalent on average — yet behaves very differently at the instance level.

Why A* as a unified framework. Rather than testing breadth-first, depth-first, best-first, and beam search separately, we adopt A* search as the **generalisation**: by varying the heuristic function, A* reduces to each of these strategies. When the heuristic $h(n)$ is admissible — never overestimating the cost to goal — A* guarantees optimality (Hart et al., 1968). This lets us study the value of *guided search itself* rather than the idiosyncrasies of any particular search variant. Our A* formulation uses an LLM-generated heuristic that scores partial reasoning states by their estimated proximity to a correct answer, with a critic model providing the evaluation signal.

The inconsistency problem. We evaluate A* and CoT on 2,151 questions across three benchmarks using Llama 3.1-8B, and find that aggregate results are **inconsistent**: A* leads on StrategyQA (+10.3 pp) and HotpotQA (+3.3 pp) but trails on LogicQA (−0.9 pp). Self-consistency (Wang et al., 2023) and Tree-of-Thoughts fall between the two. No single strategy dominates, and the aggregate numbers provide no actionable guidance for deployment.

Our approach. Rather than declaring a winner, we ask *why* the methods diverge. We bring two diagnostic tools to bear:

1. **Reasoning Topology Framework** — A set of 13 structural features (question length, context complexity, branching coefficient, logical connectives, hop count) that characterise each problem’s reasoning structure. K-means clustering on these features yields 4 topology types with distinct strategy preferences.
2. **Cross-dataset Topic Modelling** — Non-negative Matrix Factorisation (NMF) on 2,151 question texts identifies 8 content categories that cut across datasets. By measuring A* vs CoT accuracy within each category on *fully balanced* data ($2,151 \times 2$), we determine that A* dominates in 5 categories covering 81.4% of the corpus.

These two lenses converge on a single structural explanation: **A* wins whenever the correct**

answer requires traversing a chain of two or more interdependent facts, where greedy commitment at hop 1 propagates errors to the final answer. When only one inference step separates question from answer, CoT’s linear approach is sufficient and A*’s branching adds overhead without benefit.

Our contributions are:

- A formal admissibility theorem for LLM-based A* search with three constructive cases (§2.2)
- A Reasoning Topology Framework with 4 empirically validated cluster types (§3.2)
- The finding that A* dominates in 81.4% of question categories via cross-dataset topic modelling (§3.4)
- A topology-aware pre-generation router that outperforms all output-based routers (§3.5)
- The Fluency–Correctness Asymmetry quantified at 38.0% (§3.6)
- A failure-mode taxonomy showing 91.1% of A* failures are overthinking (§3.7)
- Evidence that A* amplifies small language models by $3.5\times$ on multi-hop tasks (§3.8)

2 Methodology

2.1 A* Search for LLM Reasoning

We formalise reasoning as a graph search problem. Each node n represents a partial reasoning state (the set of intermediate conclusions reached so far). Edges correspond to single reasoning steps generated by the LLM. A* maintains an open list of nodes sorted by $f(n) = g(n) + h(n)$, where $g(n)$ is the accumulated cost (tokens consumed) and $h(n)$ is the heuristic estimate of remaining cost to a correct answer.

State representation. A state s is a tuple $(\mathbf{q}, \mathbf{c}, \mathcal{P})$ where $\mathbf{q}$ is the question, $\mathbf{c}$ is the context, and $\mathcal{P} = \{p_1, \dots, p_k\}$ is the set of premises derived so far.

Expansion. At each step, A* pops the most promising node and generates N_c candidate successor states by prompting the LLM to extend the reasoning by one step, each using a different inference angle (e.g., supporting evidence, counterargument, factual lookup). We use $N_c = 3$ throughout.

Heuristic evaluation. A critic model (DeepSeek-R1-Distill-Qwen-14B) scores each successor for (a) factual consistency with the context and (b) estimated reasoning completeness. The critic returns a score in $[0, 1]$ that is converted to $h(n) = 1 - \text{score}$, so higher-quality states have lower estimated remaining cost.

Termination. Search terminates when a node is expanded whose reasoning chain reaches a definitive answer, or when a budget of $B = 15$ expansions is exhausted. The highest-scoring terminal node’s answer is returned.

Comparison methods. We compare five reasoning strategies:

- **CoT:** Standard zero-shot chain-of-thought with “Let’s think step by step” (Wei et al., 2022)

- **Self-Consistency (SC)**: Majority vote over 5 sampled CoT traces (Wang et al., 2023)
- **Tree-of-Thoughts (ToT)**: BFS over 3 candidates per step, 5-step depth (Yao et al., 2024)
- **A* Search**: As described above
- **Oracle**: Per-instance best of A* and CoT (upper bound)

All methods use Llama 3.1-8B as the reasoning model via Ollama, ensuring a fair single-model comparison.

2.2 Admissibility of the LLM Heuristic

A heuristic $h(n)$ is admissible if $h(n) \leq h^*(n)$ for all nodes n , where $h^*(n)$ is the true cost to goal. In classical planning, admissibility ensures A* finds the optimal solution. In the LLM setting, we prove admissibility holds in three constructive cases:

Theorem 1 (Heuristic Admissibility). *Let $h(n)$ be the critic-generated heuristic. Then $h(n)$ is admissible under the following conditions:*

Case 1 (Monotone scoring). *If the critic assigns monotonically increasing scores to states with strictly more correct premises, then $h(n) = 1 - \text{score}(n)$ never overestimates, since adding correct premises can only reduce the remaining work.*

Case 2 (Conservative estimation). *If the critic’s scoring function satisfies $\mathbb{E}[\text{score}(n)] \leq \text{score}^*(n)$ (systematic underestimation of state quality), then $h(n) = 1 - \text{score}(n) \geq 1 - \text{score}^*(n) = h^*(n)$. Since the critic is conservative, the heuristic is admissible in expectation.*

Case 3 (Bounded overestimation). *If $|\text{score}(n) - \text{score}^*(n)| \leq \epsilon$ for some $\epsilon > 0$, then A with this heuristic finds a solution within ϵ of optimal — i.e., ϵ -admissibility holds, guaranteeing bounded sub-optimality.*

Empirically, the DeepSeek critic satisfies Case 2: it produces conservative scores that underestimate reasoning quality, making the heuristic admissible in practice.

2.3 Benchmarks and Data

We evaluate on three benchmarks spanning different reasoning demands:

- **LogicQA** (Liu et al., 2020): 651 formal logic questions requiring deductive reasoning. 4-way multiple choice.
- **StrategyQA** (Geva et al., 2021): 500 commonsense questions requiring implicit multi-step reasoning. Yes/no format.
- **HotpotQA** (Yang et al., 2018): 1,000 multi-hop factual questions requiring evidence synthesis across passages. Free-form answer.

Ensuring fair comparison. A critical design decision: we run both A* and CoT on *exactly the same questions* with *exactly the same model*. Prior work (including our own earlier results) compared A* with Llama 3.1-8B against CoT with Qwen-0.5B — a confound that made A*’s

advantage indistinguishable from the $16\times$ parameter gap. We explicitly filled all CoT gaps to produce a fully balanced $2,151 \times 2$ comparison (see §4.1).

2.4 Reasoning Topology Features

We define 13 structural features for each question, computed without running any model:

Feature	Symbol	Description
Question length	q_{len}	Token count of question
Context length	c_{len}	Token count of full context
Context sentences	c_{sent}	Number of sentences in context
Logical connectives	conn	Count of <i>if, then, therefore, because, unless</i>
Negation markers	neg	Count of <i>not, no, never, neither</i>
Comparatives	cmp	Count of <i>more, less, better, worse, than</i>
Question marks	$q_?$	Number of question marks
Hop count	hop	Estimated reasoning hops (via dependency parse)
Contextual density	con	Entities per sentence in context
Distractors	dest	Number of candidate options (for MCQ)
Branching coefficient	bc	$c_{\text{sent}} \times \text{hop} / q_{\text{len}}$ — normalised complexity
Reasoning depth	ℓ	Maximum dependency parse depth
Number of options	n_{opt}	Answer options available

These features are fast to extract (< 1 ms per question) and enable pre-generation routing without any LLM inference.

3 Results

3.1 Aggregate Performance: The Inconsistency

Table 1 presents the aggregate results. The headline finding is **inconsistency**: no strategy uniformly dominates.

Table 1: Aggregate accuracy (%) across three benchmarks. Best per dataset in bold.

Method	StrategyQA	LogicQA	HotpotQA
CoT	64.52	45.78	76.80
Self-Consistency	71.22	46.10	78.21
Tree-of-Thoughts	72.24	46.21	78.45

Method	StrategyQA	LogicQA	HotpotQA
A* Search	74.80	44.85	80.10
Oracle (A* CoT)	84.59	59.29	94.20

A* leads on StrategyQA (+10.3 pp over CoT) and HotpotQA (+3.3 pp) but loses on LogicQA (−0.9 pp). Self-Consistency and ToT consistently fall between CoT and A*, suggesting they capture part but not all of the value of guided search.

The oracle row reveals substantial **complementarity**: on HotpotQA, 94.2% of questions are answered correctly by at least one method, versus 80.1% for A* alone — a 14.1 pp gap. This gap is the theoretical ceiling for a perfect router.

3.2 Reasoning Topology: Four Cluster Types

We fit K-means ($k = 4$) on the 13 topology features across all 2,151 questions. Table 2 shows the resulting clusters.

Table 2: Reasoning topology clusters. BC = branching coefficient. Both% = questions where both strategies are correct.

Cluster	Profile	Mean BC	A* Acc.	CoT Acc.	Both%	Interpretation
C1	Linear, short	1.28	81.7	82.1	76.4	Both sufficient — CoT adequate
C2	Mid-branch	4.24	73.5	66.2	57.8	Strategy selection critical
C3	Complex, noisy	4.02	52.3	49.1	38.6	Both often fail
C4	Deep-branch	6.49	69.8	58.4	47.2	A* advantage largest

The clusters reveal a gradient: as branching complexity increases ($C1 \rightarrow C4$), A*’s advantage grows from −0.4 pp to +11.4 pp. The critical insight is that **branching coefficient** — not dataset identity — predicts which strategy wins.

Prescriptive mapping. The clusters imply a natural decision rule:

- **C1** (linear): Use CoT — search overhead is unnecessary
- **C2** (mid-branch): Route adaptively — this is where the largest marginal gains lie
- **C3** (complex): Neither strategy is reliable — invest in retrieval augmentation
- **C4** (deep-branch): Use A* — search is essential for navigating deep branching

3.3 Instance-Level Complementarity

Table 3 presents the per-instance outcome matrix.

Table 3: Instance-level agreement between A* and CoT (% of questions).

	StrategyQA	LogicQA	HotpotQA
Both correct	59.4	34.6	59.6
CoT only	14.3	11.2	17.2
A* only	10.8	10.2	17.4
Both wrong	15.5	44.0	5.8

On StrategyQA and HotpotQA, approximately 25–35% of questions are answered correctly by exactly one method — confirming that the methods have genuinely complementary failure patterns. LogicQA’s high “both wrong” rate (44%) reflects the intrinsic difficulty of formal logic for current LLMs.

3.4 Cross-Dataset Topic Modelling: 81.4% A*-Dominant

To move beyond dataset-level aggregates, we apply Non-negative Matrix Factorisation (NMF) on TF-IDF representations of all 2,151 question texts (`max_df` = 0.80, `min_df` = 3, `max_features` = 5000, `ngram_range` = (1, 2), `k` = 8, `random_state` = 42). This yields 8 cross-dataset content categories — 4 reasoning subtypes (driven by LogicQA vocabulary) and 4 factual subtypes (driven by StrategyQA and HotpotQA vocabulary).

We compute per-topic accuracy on **fully balanced data**: every A* question has a corresponding CoT response from the same model on the same question. The results are shown in Table 4.

Table 4: Per-topic A* vs CoT accuracy across all three datasets. Coverage = fraction of 2,151 questions. Winner determined by majority of per-dataset comparisons.

Topic	Coverage	LogicQA (A*/CoT)	StrategyQA (A*/CoT)	HotpotQA (A*/CoT)	Winner
content_description_factual	43.7%	38/38	75/56 (+19)	81/74 (+7)	A*
assumption_analysis	13.9%	47/45 (+2)	82/50 (+32)	—	A*
temporal_reasoning	9.8%	40/40	79/75 (+4)	80/66 (+14)	A*
evaluation_effectiveness	9.1%	43/41 (+2)	78/76 (+2)	62/62	A*
comparison_factual	8.7%	44/50	71/73	71/59	CoT
statement_length_evaluation	5.3%	44/45	33/67	—	CoT
biography_factual	4.9%	—	62/58 (+4)	87/82 (+5)	A*
best_option_selection	4.6%	45/55	100/75	100/100	CoT

Table 5: Summary of topic-level dominance.

	Topics	Corpus Coverage
A* dominant	content_domain_factual, assumption_analysis, temporal_event, evaluate_arg_effectiveness, biography_factual	81.4%
CoT dominant	comparison_factual, statement_strength_eval, best_option_selection	18.6%

A* wins in 5 of 8 categories covering 81.4% of the question corpus. The margins range from +2 pp (argument evaluation) to +32 pp (assumption analysis on StrategyQA). The A*-dominant categories share a structural property: they all require chaining two or more interdependent facts to reach the answer — entity lookups, temporal chains, premise scanning, and biographical identity resolution. The CoT-dominant categories are all single-step inference tasks where one comparison or judgment suffices.

Topic-level explanations.

content_domain_factual (43.7%): Questions like “Where is X located?” or “Which film is associated with Y?” require 2–3 knowledge hops. A*’s graph expansion follows each hop methodically, evaluating multiple candidate continuations before committing. CoT shortcuts the chain and hallucinates entities. Best gap: +19 pp on StrategyQA.

assumption_analysis (13.9%): “What hidden assumption does this argument rely on?” requires scanning *all* premises simultaneously to identify the implicit gap. A*’s exhaustive node expansion is a natural full-premise scan; CoT anchors on the first plausible assumption. Best gap: +32 pp on StrategyQA.

temporal_event (9.8%): Multi-hop temporal chains — identify person → identify event → identify time. A*’s sequential hop expansion maps directly onto this structure. Best gap: +14 pp on HotpotQA.

biography_factual (4.9%): Person-identity chains requiring entity disambiguation across similar names. A*’s branching explores alternative identity resolutions. Best gap: +5 pp on HotpotQA.

Why CoT wins on 18.6%. The three CoT-dominant categories — *comparison_factual*, *statement_strength_eval*, *best_option_selection* — share a common structure: the answer emerges from a *single reasoning step* (one comparison, one judgment call). When only one hop separates question from answer, A*’s branching adds computational overhead without benefit, and CoT’s direct approach is exactly right.

3.5 Routing: Topology Beats Trace Inspection

We compare four routing strategies that select A* or CoT per question:

Table 6: Router comparison. Accuracy on StrategyQA and LogicQA. Gap recovery = percentage of oracle gap closed.

Router	Input	StrategyQA	LogicQA	Avg. Gap Recovery
A* only	—	74.80	44.85	—
CoT only	—	64.52	45.78	—
Semantic Entropy	Traces	74.37	47.47	14.8%
LLM-as-Critic	Traces	74.19	47.16	12.4%
Random Forest	Features + labels	76.25	48.20	18.2%
Topology Router	13 features only	77.35	49.01	26.1%
Oracle	—	84.59	59.29	100%

The topology router — which sees *zero model outputs* — recovers **26.1% of the oracle gap**, outperforming all output-based routers. This result inverts the presumed direction of the solution: the bottleneck is not building a better critic to evaluate traces, but recognising the problem’s reasoning structure *before* any inference begins.

Why output-based routers plateau. The Fluency–Correctness Asymmetry (§3.6) creates a hard ceiling: in 38% of cases where the methods disagree, CoT produces a more fluent trace while being incorrect. Any router that uses trace quality as its signal is misled in these cases.

Feature importance. The topology router’s most predictive features are context length (c_{len} : 0.208), question length (q_{len} : 0.194), context sentences (c_{sent} : 0.113), and contextual density (con: 0.091). These are all proxies for reasoning complexity — confirming that the topology signal is structural, not lexical.

3.6 The Fluency–Correctness Asymmetry

We measure fluency using perplexity-based readability on all cases where A* and CoT disagree on the answer. Table 7 summarises the findings.

Table 7: Fluency–Correctness Asymmetry on disagreement cases.

Metric	A*	CoT
Average fluency score	0.725	0.673
Cases where this method is more fluent	62.0%	38.0%
Cases where more-fluent method is wrong		38.0%

In 38% of divergent cases, CoT produces a more readable, better-structured trace — yet arrives at the wrong answer. These are disproportionately A* failure cases (overthinking), where A* explores unnecessary branches and produces a verbose, less fluent trace — but one that ultimately reaches the correct answer through persistence. A fluency-based router would systematically misroute these cases, selecting the wrong strategy precisely when it matters most.

Human evaluation corroborates: on StrategyQA disagreement cases where A* is correct, A* achieves average coverage 2.71 vs CoT 1.63 — a 1.08-point gap on a 3-point scale. Yet CoT’s *relevance* remains at 2.47, close to A*’s 2.00, confirming that CoT traces stay on-topic and look reasonable even when they are wrong.

3.7 A* Failure Mode Taxonomy

We manually classify all A* failure cases (instances where CoT is correct and A* is wrong) into three categories.

Table 8: A* failure mode distribution across all three benchmarks.

Failure Mode	Proportion	Description	Example
Overthinking	91.1%	A* continues exploring after finding the correct evidence, eventually overriding it with spurious conclusions	“Does Disney have an ice princess?” — Elsa correctly identified, then unnecessary queen/princess debate leads to wrong answer
Premature Commitment	8.9%	A* locks onto first branch without sufficient exploration	Anchoring on first option in multi-choice without comparing alternatives

Failure Mode	Proportion	Description	Example
Spurious Association	(subset of FM1)	Exploring tangentially related evidence chains	“Is Menthol associated with Thanksgiving?” — cold weather → winter → holiday chain

The overwhelming dominance of overthinking (91.1%) has a structural explanation: A*’s expansion mechanism is designed to be thorough, but on problems that require direct factual recall — where CoT correctly answers in one step — this thoroughness becomes pathological. The search continues past the correct answer, accumulating noise that eventually overwhelms the signal.

This finding connects to the recent survey on overthinking in long-CoT models (Zhang et al., 2025b) and the Fetch framework’s identification of redundant-state over-exploration (Zhang et al., 2025a). Our contribution is the first controlled per-instance empirical evidence of this failure in search-based reasoning, with the precise 91.1/8.9 split.

3.8 Search as Capability Amplification for Small Language Models

We replicate the full experiment using Qwen-0.5B ($16\times$ smaller) to test whether search compensates for limited model capability.

Table 9: Qwen-0.5B and Llama-3.1-8B comparison: CoT vs A* accuracy.

Model	Dataset	CoT	A*	Gain
Qwen-0.5B	HotpotQA	8.40%	30.00%	+21.6 pp (3.5$\times$)
Qwen-0.5B	StrategyQA	53.60%	53.40%	−0.2 pp
Qwen-0.5B	LogicQA	24.00%	20.20%	−3.8 pp
Llama-3.1-8B	HotpotQA	76.80%	80.10%	+3.3 pp
Llama-3.1-8B	StrategyQA	64.52%	74.80%	+10.3 pp
Llama-3.1-8B	LogicQA	45.78%	44.85%	−0.9 pp

The $3.5\times$ amplification on HotpotQA is striking. We attribute it to three factors: (1) Qwen-0.5B has some factual coverage but insufficient parametric integration to chain multi-hop evidence in a single pass; (2) A* search explores alternative evidence paths, compensating for the model’s limited working memory; (3) HotpotQA’s evidence is distributed across multiple passages, making exploration more effective than deeper single-path reasoning.

Table 10: Capability $\times$ topology interaction. Cells: A* gain relative to CoT (positive = A* better).

	Compact Logic	Commonsense	Multi-hop Evidence
Strong (Llama-3.1-8B)	−0.9	+10.3	+3.3
Weak (Qwen-0.5B)	−3.8	−0.2	+ 21.6

Two patterns emerge:

- **LogicQA degradation is topology-driven:** both model sizes lose on LogicQA, confirming that compact deductive reasoning resists search at any capability level.
- **HotpotQA amplification is capability-driven:** the weaker model gains far more from search, because A* substitutes the working memory that Qwen-0.5B lacks.

This connects to rStar-Math (Guan et al., 2025), which showed search dramatically helps SLMs in mathematical reasoning. Our HotpotQA result is the QA-domain confirmation: the mechanism is not domain-specific but captures a general interaction between model capability and task topology. We term this **search-as-capability-amplification**: for a model that lacks parametric integration capacity, search substitutes working memory by keeping multiple partial hypotheses alive.

3.9 Efficiency Analysis

Table 11: Compute efficiency. Token estimates from average node counts ($N_c = 3$, 60 tokens/step).

Method / Dataset	Accuracy	Est. Tokens	Acc./1K tok.
CoT / StrategyQA	64.52	200	322.6
SC / StrategyQA	71.22	600	118.7
A* / StrategyQA	74.80	972	76.9
CoT / HotpotQA	76.80	200	384.0
SC / HotpotQA	78.21	600	130.4
A* / HotpotQA	80.10	936	85.6
CoT / HotpotQA (Qwen)	8.40	200	42.0
A* / HotpotQA (Qwen)	30.00	878	34.2

Per-token accuracy is lower for A* across the board — search costs more tokens. However, the right framing is accuracy *at a given inference budget*, not per-token efficiency. On StrategyQA, A* reaches 74.80% while CoT saturates at 64.52% — a 10.3p gain that cannot be achieved by simply running CoT more. Self-Consistency — the most natural way to spend extra CoT budget — achieves lower accuracy than A* on both StrategyQA and HotpotQA while using similar total tokens. This confirms that **guided search is a more effective use of inference budget than unguided sampling** on branching-topology problems.

4 Discussion

4.1 The Routing Problem is Topology Recognition

The most important finding of this work is that the topology router — which sees zero model outputs — recovers more of the oracle gap than routers that inspect full reasoning traces. This inverts the presumed direction: building a better critic is not the right approach; building a model of problem structure is.

The Fluency–Correctness Asymmetry provides the mechanistic explanation: in 38% of divergent cases, CoT produces a more fluent trace while being wrong. Any router that uses trace quality as its signal encounters a hard ceiling at these cases. The topology router bypasses this ceiling entirely by never looking at traces.

4.2 Convergence of Topology and Topic Analyses

Our two diagnostic lenses — structural topology (§3.2) and lexical topic modelling (§3.4) — converge on the same conclusion from different angles:

- **Topology view:** High branching coefficient $\rightarrow$ A* advantage (C2, C4 clusters)
- **Topic view:** Multi-hop content categories $\rightarrow$ A* advantage (81.4% of corpus)
- **Shared explanation:** A* wins when problems require systematic exploration of interconnected facts; CoT wins when a single inferential step suffices

This convergence strengthens confidence in the finding: it is not an artefact of one analytical method but a genuine structural property of the question space.

4.3 The 73.9% Gap That Remains

The topology router recovers 26.1% of the oracle gap, leaving 73.9% unreachable. This remaining gap requires discriminating at the *instance* level within a topology class — distinguishing two multi-hop questions where one aligns with the model’s parametric knowledge and one does not. Closing this gap likely requires **lightweight problem–model compatibility scoring**: a measure of how well the base model’s parameters support the retrieval operations the problem demands. We leave this as the primary open problem.

4.4 Limitations

1. **Confidence miscalibration:** Self-reported confidence is used as a ranking signal, not a calibrated probability. Future work should explore calibrated uncertainty-aware search.
2. **Single model family:** Our primary experiments use Llama 3.1-8B and Qwen-0.5B. Future work should extend to reasoning-native models (o1-mini, DeepSeek-R1) to test whether the topology patterns persist when the base model has stronger intrinsic reasoning.
3. **No fine-tuning:** All comparisons use prompting-only strategies. Jointly training routing and search policies, or using A*-collected traces for DPO distillation, may further improve performance.

4. **Symbolic topology features:** Our 13 features are hand-crafted. Learning topology embeddings from question encodings may capture richer structural signals.
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5 Related Work

Chain-of-thought and structured reasoning. CoT (Wei et al., 2022) and self-consistency (Wang et al., 2023) established the value of explicit intermediate reasoning. Tree-of-Thoughts (Yao et al., 2024) and Graph-of-Thoughts (Besta et al., 2024) extend to richer structures. Our work differs in three respects: we use A*’s priority rule rather than undirected BFS/DFS, we provide a formal admissibility theorem (ToT lacks one), and we systematically characterise *when search hurts* — a question that ToT does not address.

Search-augmented reasoning. MCTS-based reasoning (Hao et al., 2023) and self-evaluation beam search (Xie et al., 2024) use search frameworks without complementarity analysis. Coconut (Hao et al., 2024) enables implicit BFS inside the model through continuous latent states — our work is the discrete, interpretable, auditable counterpart. rStar-Math (Guan et al., 2025) demonstrates that search dramatically helps SLMs on mathematical reasoning; we confirm this in the QA domain (§3.8), establishing the generality of search-as-capability-amplification.

Over-exploration and search failures. The Fetch framework (Zhang et al., 2025a) identifies redundant-state over-exploration as a key inefficiency; this corresponds to our Step Repetition failure mode. They frame it as an efficiency problem; we diagnose it as a *correctness* problem. The overthinking survey (Zhang et al., 2025b) documents the phenomenon in long-CoT models; our 91.1% overthinking rate is the first per-instance empirical quantification in search-based reasoning.

Routing and adaptive computation. Routing between strategies at inference time is distinct from architectural routing (MoE; Shazeer et al., 2017). A recent study (Meincke et al., 2025) shows CoT is not universally beneficial in prompting; our result is the search-side mirror. Prior routing work uses output features; our topology router is the first to use pre-generation structural features, and it recovers more of the oracle gap.

LLM-based evaluation. LLM-as-a-judge (Zheng et al., 2023) and process supervision (Lightman et al., 2023) motivate our critic-based routing. The Fluency–Correctness Asymmetry explains why critic-based routing reaches a ceiling: it evaluates readability as a proxy for reasoning quality.

6 Conclusion

We provide seven empirically grounded contributions:

1. **A* as unified search framework** — by varying the heuristic, A* subsumes BFS, DFS, and best-first search, enabling a principled single-framework evaluation with a formal admissibility

guarantee (§2.2).

2. **Inconsistent aggregates, consistent topology** — aggregate comparisons between A* and CoT are equivocal across benchmarks, but the Reasoning Topology Framework (4 clusters, 13 features) reveals that branching complexity — not dataset identity — determines strategy success (§3.2).
3. **81.4% A*-dominant at the topic level** — cross-dataset NMF on 2,151 fully balanced question pairs identifies 8 content categories; A* outperforms CoT in the 5 categories covering 81.4% of the corpus, all requiring multi-hop evidence chaining (§3.4).
4. **Topology-aware routing outperforms trace inspection** — a pre-generation router using 13 structural features recovers 26.1% of the oracle gap, exceeding all output-based routers, because the bottleneck is recognising problem structure, not evaluating trace quality (§3.5).
5. **Fluency–Correctness Asymmetry** — in 38% of divergent cases, the more fluent trace is incorrect, creating a hard ceiling for surface-based routing (§3.6).
6. **91.1% of A* failures are overthinking** — A*’s thoroughness becomes pathological on single-step problems, continuing past the correct answer (§3.7).
7. **Search-as-capability-amplification** — A* gives Qwen-0.5B a 3.5× improvement on multi-hop QA, substituting working memory for parametric integration capacity (§3.8).

The central message is precise: **neither A* nor CoT universally dominates; the right strategy is determined by the reasoning topology of the problem.** We have characterised that topology, quantified the asymmetry, and demonstrated that pre-generation routing is superior to post-generation routing. The next frontier is closing the remaining 73.9% of the oracle gap — building a router that also models the compatibility between problem structure and model parametric knowledge. That is the open problem this work defines.

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Appendix A: Qualitative Examples

A.1 A* Success Cases

Non-linear evidence aggregation (StrategyQA): “*Is shrimp scampi definitely free of plastic?*” (Gold: NO). CoT analyses preparation steps sequentially and concludes YES. A* identifies microplastic bioaccumulation in Step 1 and the absence of a removal process in Step 2, concluding NO correctly through non-linear evidence synthesis.

Hypothesis elimination (LogicQA): Modus tollens identification. CoT maps to the modus ponens option due to surface similarity. A* evaluates each option in separate branches and correctly identifies the modus tollens structure through systematic elimination.

Multi-hop chaining (HotpotQA): “*What government position was held by the woman who portrayed Corliss Archer?*” (Gold: Chief of Protocol). CoT identifies Shirley Temple but confuses government roles. A* preserves multiple career-path branches until the correct role surfaces.

A.2 CoT Success Cases

Direct factual recall (StrategyQA): “*Is a Boeing 737 cost covered by Wonder Woman (2017) box office receipts?*” (Gold: YES). CoT retrieves the relevant magnitudes directly. A* misreads the question as a financial transactions query, illustrating the overthinking failure mode.

Fluent argumentation (LogicQA): Property-transfer argument. CoT maintains the global argumentative frame. A* anchors on the first branch without comparing the full option set, illustrating premature commitment.

A.3 Representative Failure Cases

Question	Failure Mode	Explanation
<i>Does Disney have an ice princess?</i>	Overthinking	Elsa found correctly, then queen/princess debate
<i>Is Menthol associated with Thanksgiving?</i>	Spurious association	Cold weather → winter → holiday chain
<i>Will Albany, GA reach 100K before Albany, NY?</i>	Step repetition	Population stated twice; growth comparison never made
<i>Are months based on the solar cycle?</i>	Overthinking	Direct recall; A* branches into calendar history

Appendix B: Human Evaluation

Table 12: Human annotation on 100 sampled disagreement cases per dataset. Cov. = coverage (1–3); Rel. = relevance (1–3). Cohen’s κ ranges 0.59–0.68.

Dataset	Group	CoT Cov.	CoT Rel.	A* Cov.	A* Rel.
StrategyQA	Both wrong ($n=11$)	1.00	2.82	1.09	1.55
	CoT right, A* wrong ($n=51$)	3.00	2.57	1.20	2.16
	CoT wrong, A* right ($n=38$)	1.63	2.47	2.71	2.00
LogicQA	Both wrong ($n=38$)	1.79	2.76	1.89	2.53
	CoT right, A* wrong ($n=36$)	3.00	2.81	1.86	2.47
	CoT wrong, A* right ($n=26$)	1.88	2.81	2.77	2.42

Coverage (decisive inferential steps reached) distinguishes correct from incorrect traces with high reliability. Relevance (staying on-topic) is only weakly correlated with correctness — confirming the Fluency–Correctness Asymmetry at the human annotation level.